Circular Economy Information Systems

CEIS — live demo

Alessandro Giugno & Simeon Pilz
Institute of Computer Science, University of St.Gallen

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a.k.a CEIS

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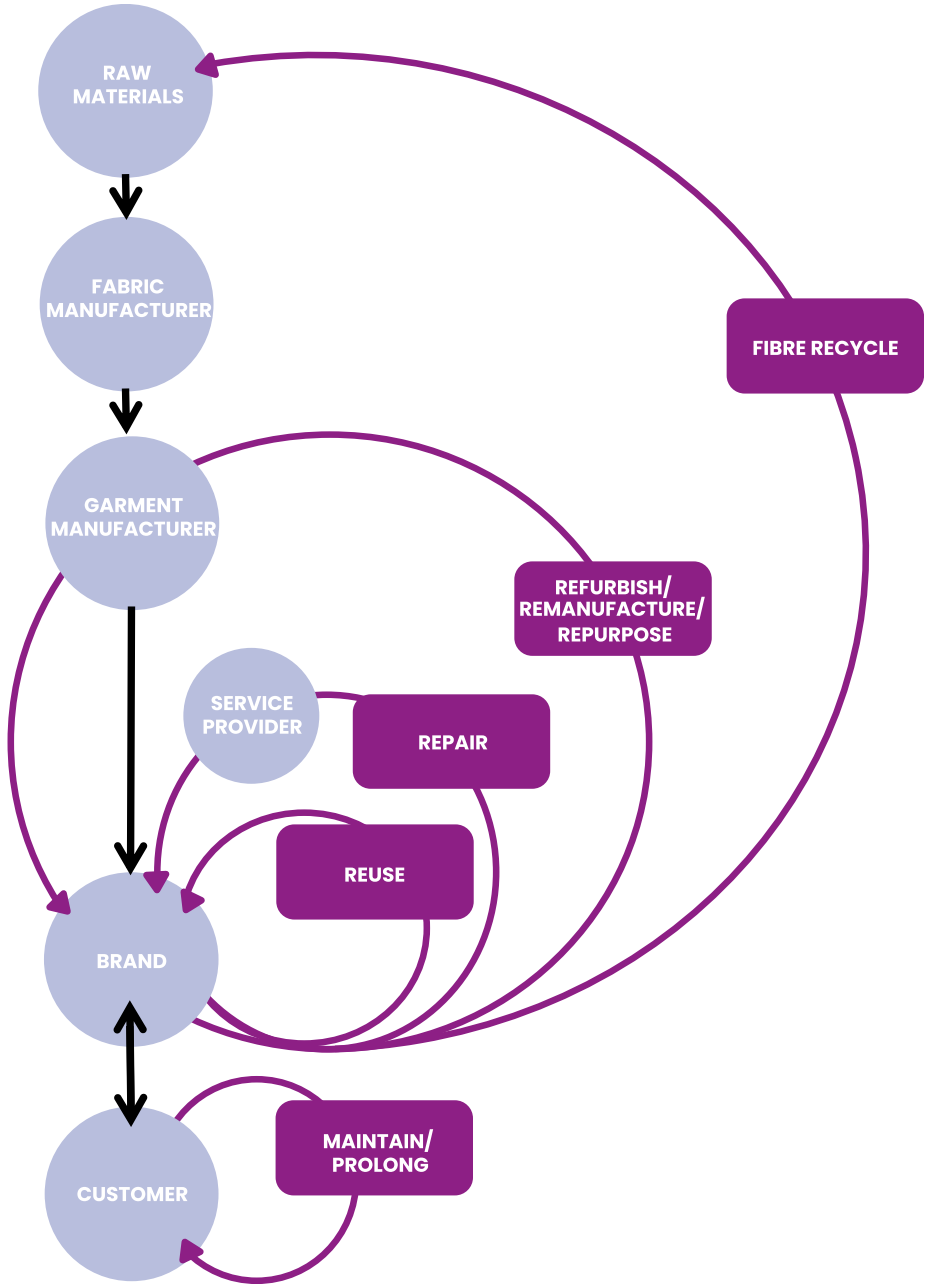
a.k.a "chase"

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From Refashion to CEIS

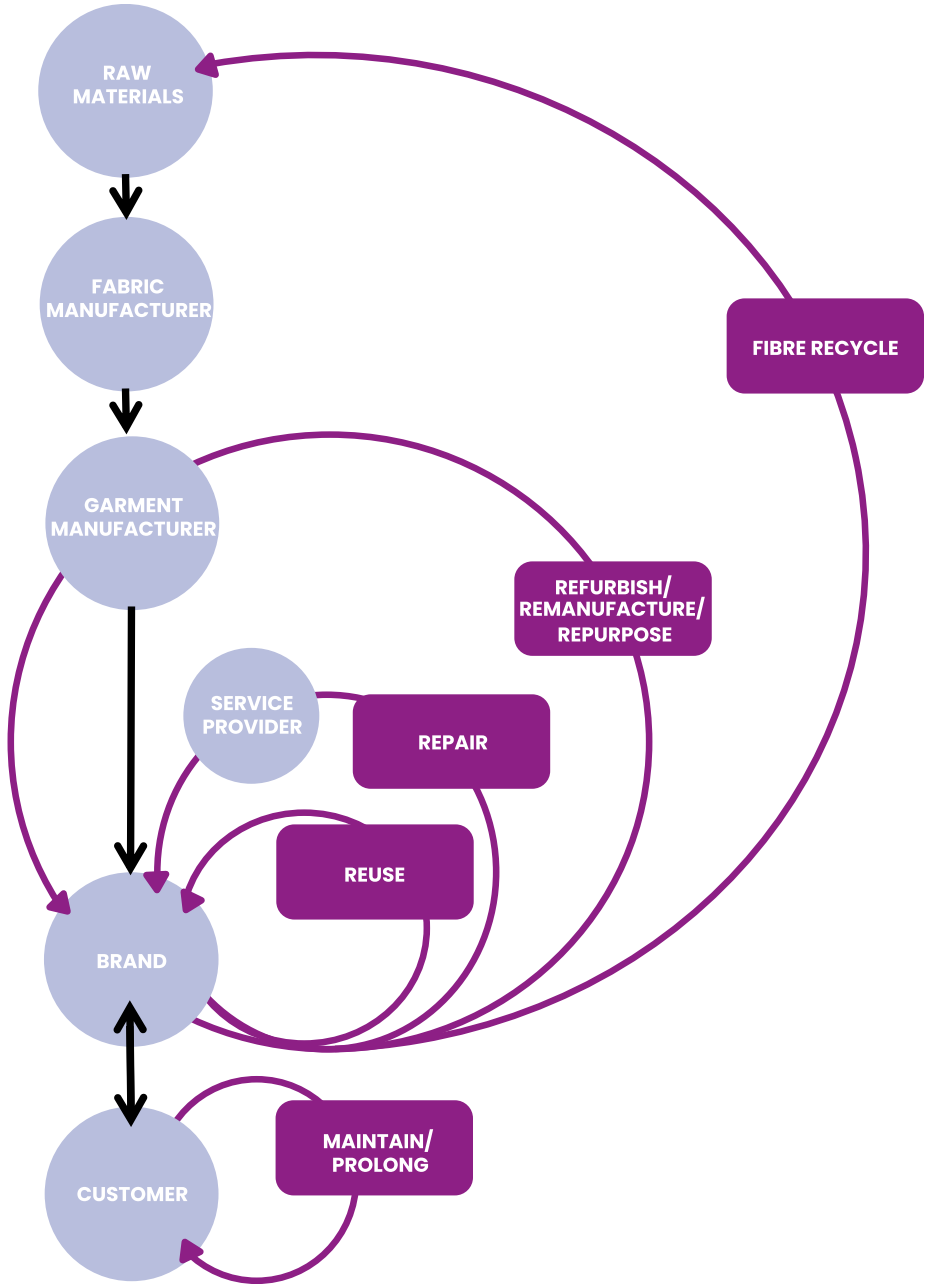




From Refashion to CEIS



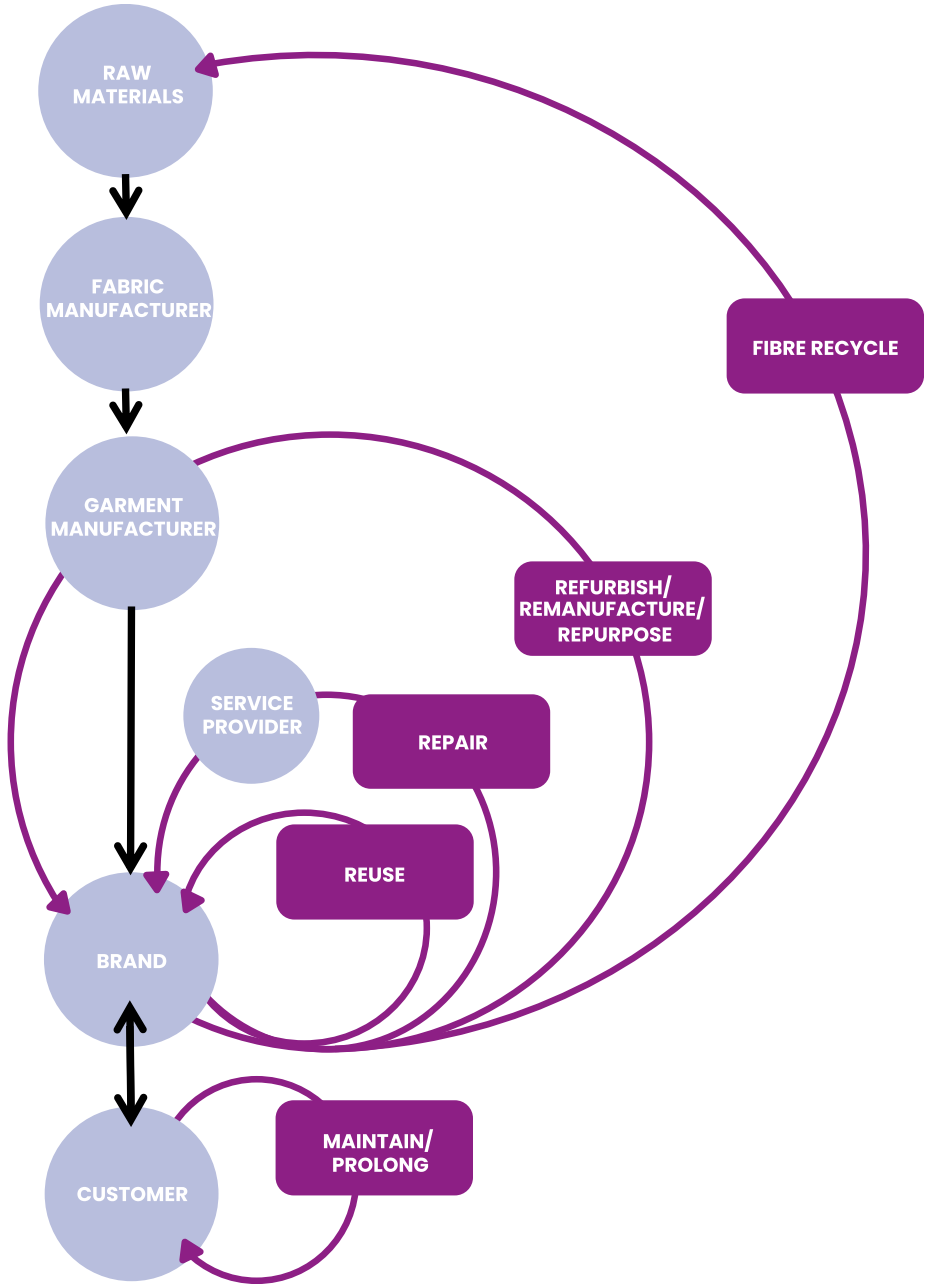
- at each stage of the product life cycle



From Refashion to CEIS



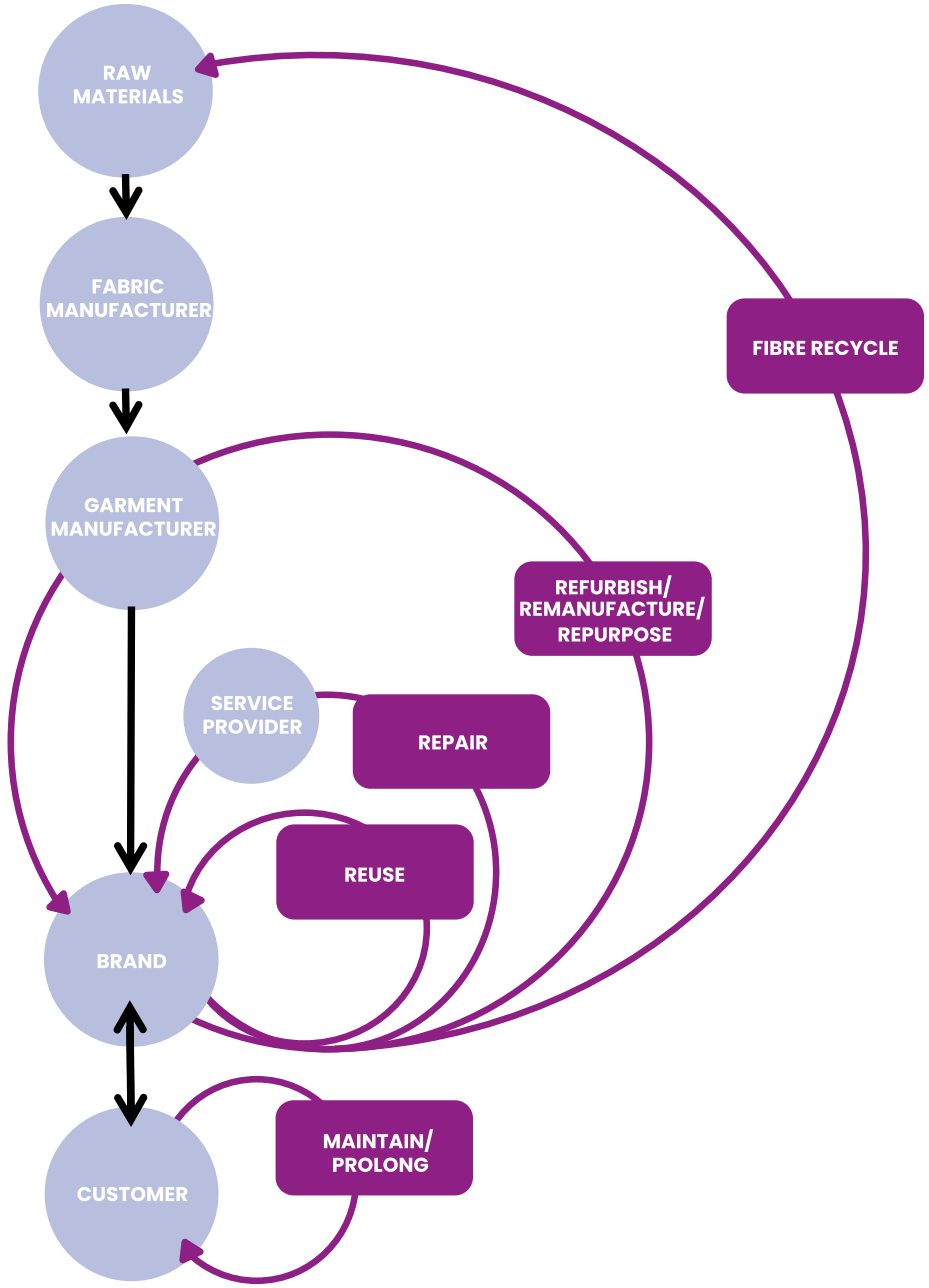
- at each stage of the product life cycle
- show to producers and consumers



From Refashion to CEIS



- at each stage of the product life cycle
- show to producers and consumers
- how their choices impact the environment

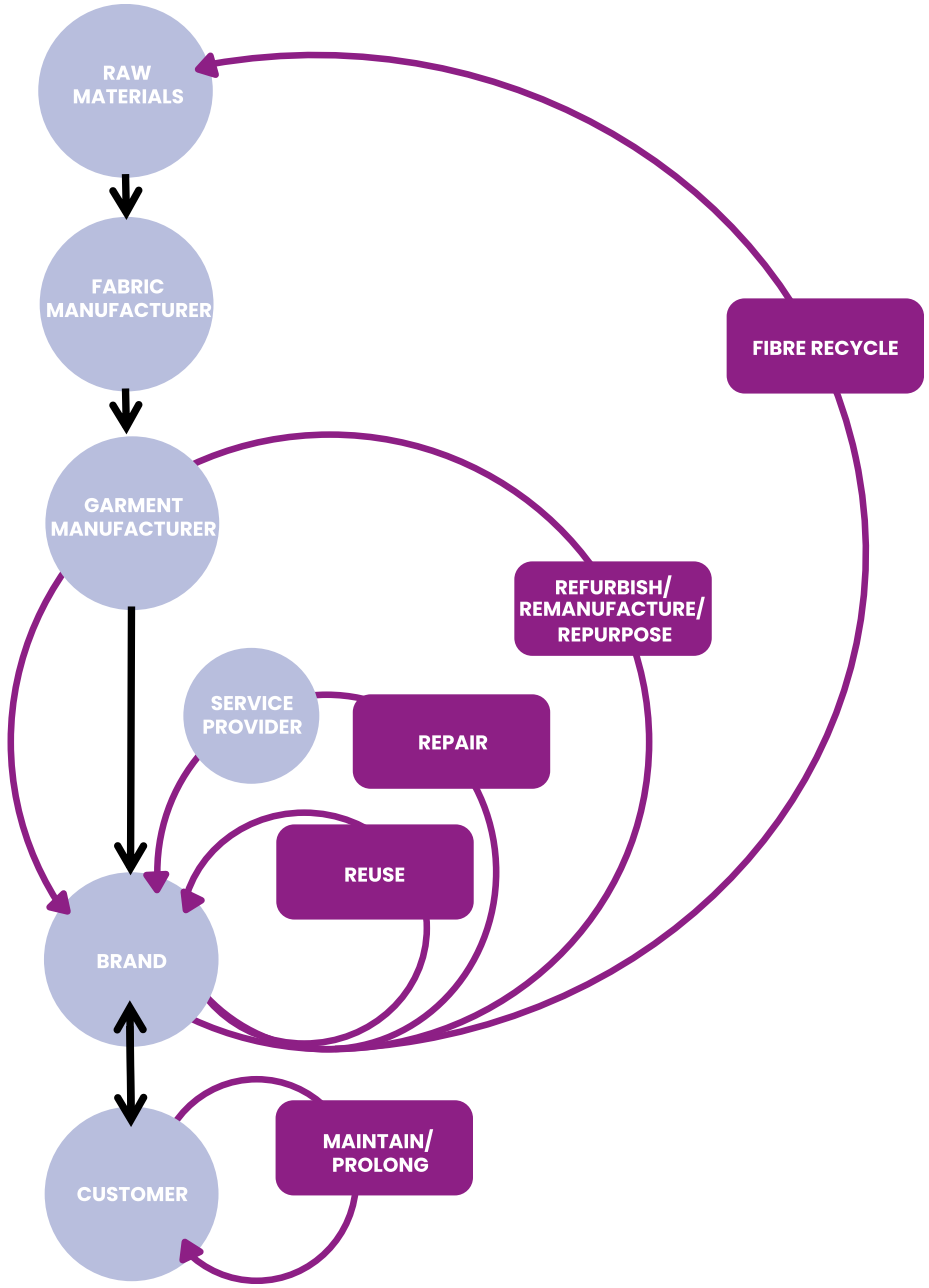


From Refashion to CEIS



- at each stage of the product life cycle
- show to producers and consumers
- how their choices impact the environment

CEIS aims to enable **informed decision-making** based on data for both **brand and customer**.



From Refashion to CEIS



- at each stage of the product life cycle
- show to producers and consumers
- how their choices impact the environment

R0
Refuse
Consumer choice

R2
Reduce
Consumer choice

CEIS aims to enable **informed decision-making** based on data for both **brand and customer**.

Numbering: https://www.circularise.com/blogs/r-strategies-for-a-circular-economy
Interpretation: Reike et al. (2018), https://doi.org/10.1016/j.resconrec.2017.08.027

From Refashion to CEIS



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R0
Refuse
Consumer choice

R2
Reduce
Consumer choice

R3
Reuse
Reverse logistics

R4
Repair
Reverse logistics

R5
Refurbish
Reverse Logistics

R6
Remanufacture
Reverse logistics

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R6
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R8
Recycle
End of life - planned

R9
Recover
End of life - planned

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From Refashion to CEIS



- at each stage of the product life cycle
- show to producers and consumers
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CEIS aims to enable **informed decision-making** based on data for both **brand and customer**.

R0
Refuse
Consumer choice

R1
Rethink
—

R2
Reduce
Consumer choice

R3
Reuse
Reverse logistics

R4
Repair
Reverse logistics

R5
Refurbish
Reverse Logistics

R6
Remanufacture
Reverse logistics

R7
Repurpose
—

R8
Recycle
End of life - planned

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Recover
End of life - planned

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Background information

Data used

- **Materials and processes:**
  - data from SOLVE Studio (Cristina Dan)
- **Emission factors:**
  - retrieved from the WISER ecosystem (ecoinvent 3.12, cut-off)
  - WISER is a digital ecosystem that simplifies access to greenhouse gas (GHG) knowledge
- **Customers and suppliers:**
  - mocked, with manufacturers and repair services around Lake Constance under alias names
    - transport distances follow these road routes

Simulated supply chain



WISER: https://wiser-climate.com/

Background information

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Simulated supply chain



Disclaimer: All CO2eq values shown in this demonstration serve illustrative purposes only.

1 - Consumer Choice

R0 Refuse R2 Reduce

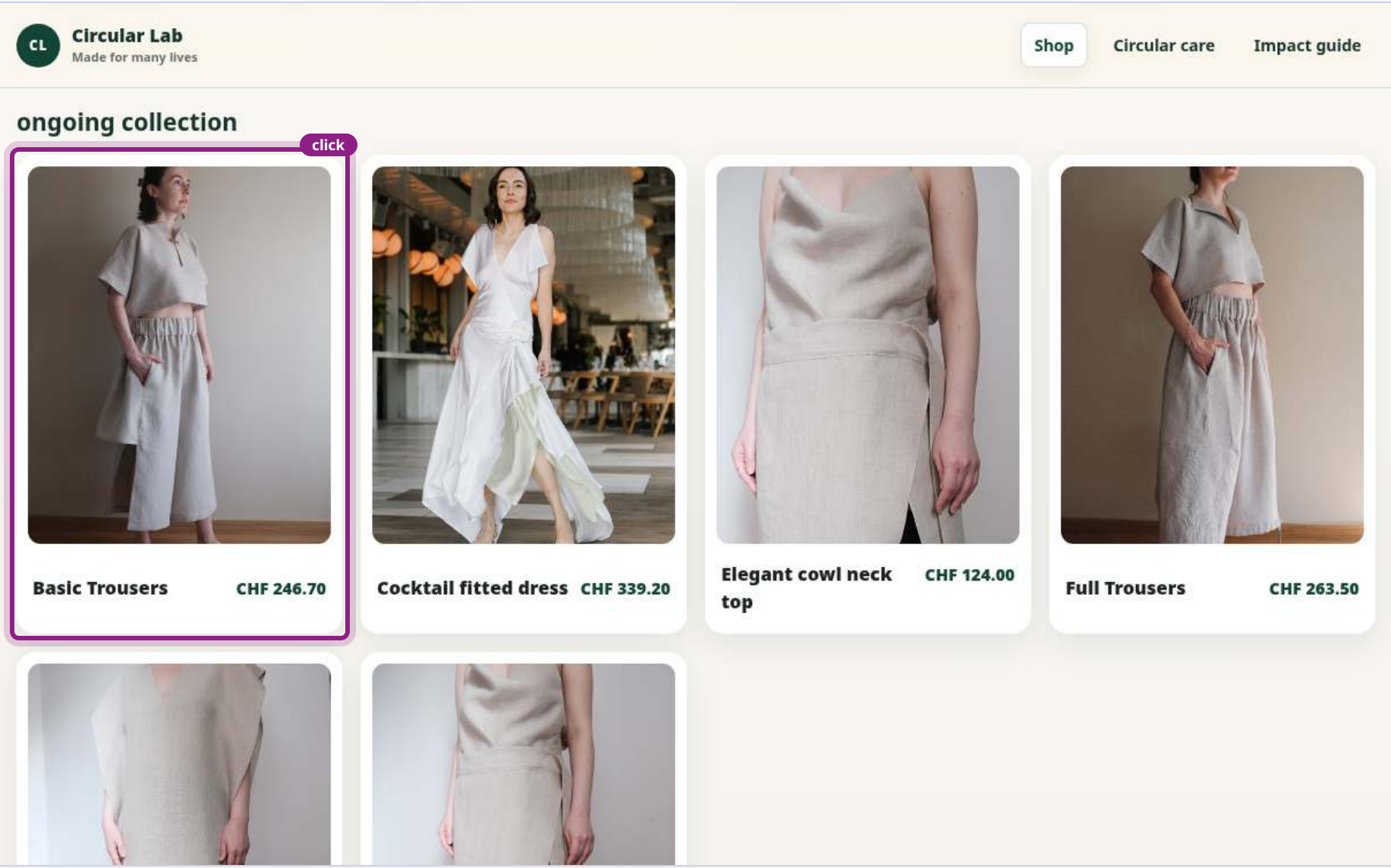
Mia wants to buy a garment. How do her choices impact the environment?

Circular Lab Shop

All mentioned prices are fictional

1 In a web shop built on CEIS, Mia browses the collection and opens the **Basic Trousers**.

http://localhost:8050/



Circular Lab Shop

All mentioned prices are fictional

2 She chooses **cotton**: the estimated impact of the trousers is **3.045 kg CO2eq**.

http://localhost:8050/garment/1

CL **Circular Lab**
Made for many lives

Shop | Circular care | Impact guide



GARMENT PRICE

CHF 246.70

Second-life materials may unlock a lower price after you choose a material.

Choose a material

select

cotton

We will immediately show available alternatives and savings.

No second-life fabric available right now

You can still order this garment in your selected material.

SELECTED MATERIAL
cotton

ESTIMATED IMPACT
3.045 kg CO2eq

Order garment

HOW IT IS MADE

Cut efficiently, built for longevity

Circular Lab Shop

All mentioned prices are fictional

3 Switching to **hemp** cuts it to **1.312 kg CO2eq**: she refuses cotton and reduces her impact.

http://localhost:8050/garment/1

CL **Circular Lab**
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Shop | Circular care | Impact guide



GARMENT PRICE

CHF 246.70

Second-life materials may unlock a lower price after you choose a material.

Choose a material

select

hemp

We will immediately show available alternatives and savings.

No second-life fabric available right now

You can still order this garment in your selected material.

SELECTED MATERIAL
hemp

ESTIMATED IMPACT
1.312 kg CO2eq

Order garment

HOW IT IS MADE

Cut efficiently, built for longevity

Circular Lab Shop

4 She orders the hemp trousers. They are not in stock, so the shop requests their production.

http://localhost:8050/garment/1

CL **Circular Lab**
Made for many lives

Shop | Circular care | Impact guide



Choose a material

hemp

We will immediately show available alternatives and savings.

No second-life fabric available right now

You can still order this garment in your selected material.

SELECTED MATERIAL
hemp

ESTIMATED IMPACT
1.312 kg CO2eq

click

Order garment

Order confirmed
Not in stock — production has been requested.

HOW IT IS MADE
Cut efficiently, built for longevity
This garment is assembled from 8 fabric pieces across 3 cut shapes.

Confirm the order

5 In the brand's Circular Lab Cockpit, Mia's order **#11** is on top as *Production requested*: **Ship**.

http://localhost:8051/

Circular Lab Cockpit | Strategy | Designer | Add Recipe | Scenario Planner

Orders

Ship an order after production is ready. Production orders create a sold garment inventory record when shipped.

ORDER	GARMENT	MATERIAL	FULFILLMENT	STATUS	CREATED	ACTION
#11	Basic Trousers	hemp	Production	Production requested	2026-11-09 11:44:35	click Ship
#10	Nordlys Dress	cotton	Stock	Completed	2026-10-23 11:08:30	Shipped
#9	Orka jacket	cotton	Production	Production requested	2026-10-23 11:05:03	Ship
#8	Elegant cowl neck top	silk	Production	Production requested	2026-10-23 11:04:38	Ship
#7	Basic Trousers	hemp	Production	Completed	2026-10-23 11:04:21	Shipped
#6	Nordlys Dress	cotton	Production	Completed	2026-10-23 11:02:14	Shipped

Pre-loved Garments

Disassemble a pre-loved garment to release all of its fabric blocks into second-hand inventory.

GARMENT ID	GARMENT	ORIGINAL GARMENT ID	FABRIC BLOCKS	DAMAGE	STATUS	ACTION

Confirm the order

6 The order is completed, and Mia's trousers are recorded as sold garment **#20**.

http://localhost:8051/

Circular Lab Cockpit | Strategy | Designer | Add Recipe | Scenario Planner

Orders

Ship an order after production is ready. Production orders create a sold garment inventory record when shipped.

ORDER	GARMENT	MATERIAL	FULFILLMENT	STATUS	CREATED	ACTION
#11	Basic Trousers	hemp	Production	Completed	2026-11-09 11:44:35	Shipped
#10	Nordlys Dress	cotton	Stock	Completed	2026-10-23 11:08:30	Shipped
#9	Orka jacket	cotton	Production	Production requested	2026-10-23 11:05:03	Ship
#8	Elegant cowl neck top	silk	Production	Production requested	2026-10-23 11:04:38	Ship
#7	Basic Trousers	hemp	Production	Completed	2026-10-23 11:04:21	Shipped
#6	Nordlys Dress	cotton	Production	Completed	2026-10-23 11:02:14	Shipped

Order #11 shipped. Garment #20 is recorded as sold.

Pre-loved Garments

Disassemble a pre-loved garment to release all of its fabric blocks into second-hand inventory.

2 - Reverse Logistics

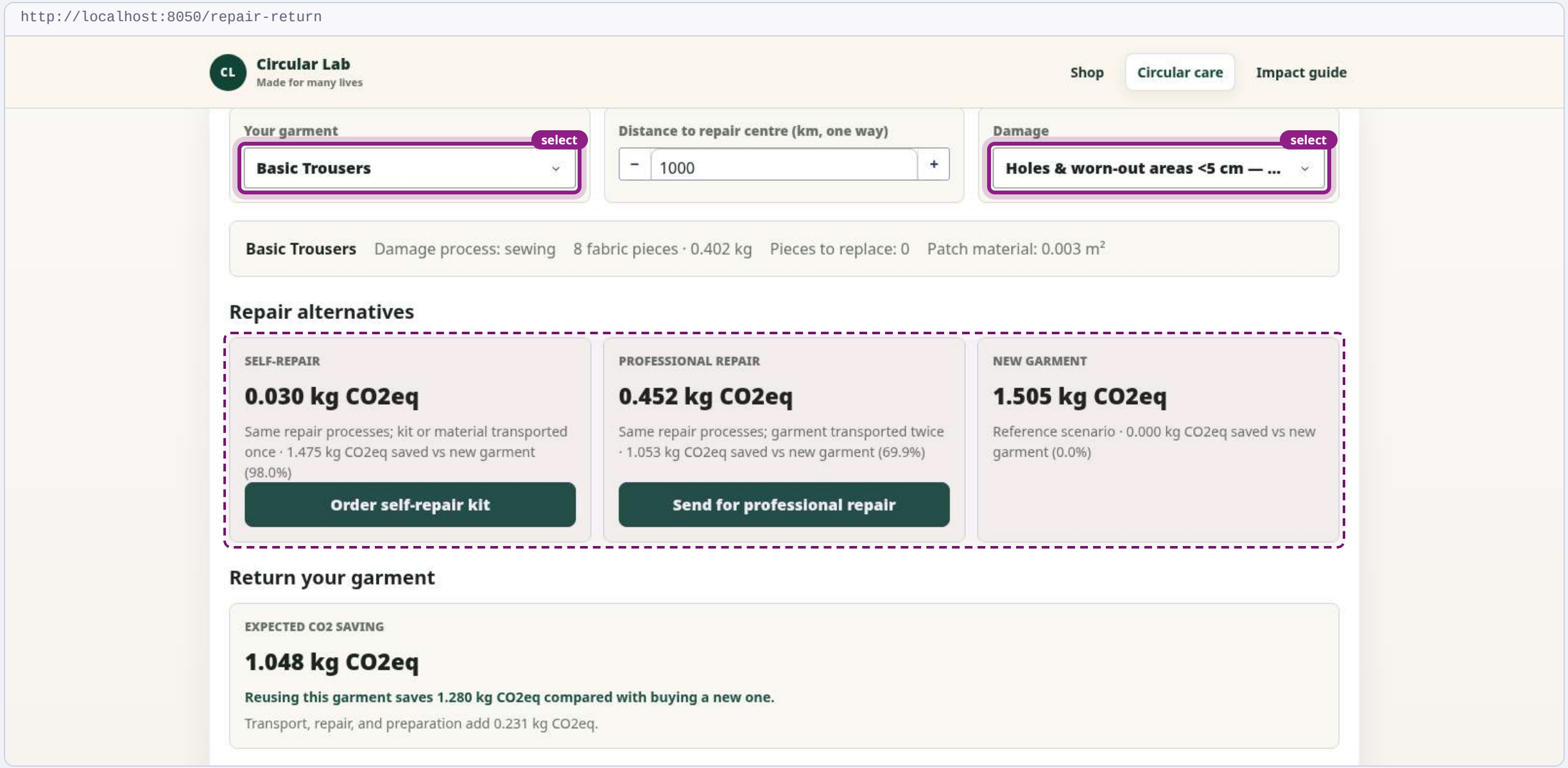
R3 Reuse | R4 Repair | R5 Refurbish | R6 Remanufacture

After some time, the condition of Mia’s garment is not satisfying anymore - **what now?**

Circular care

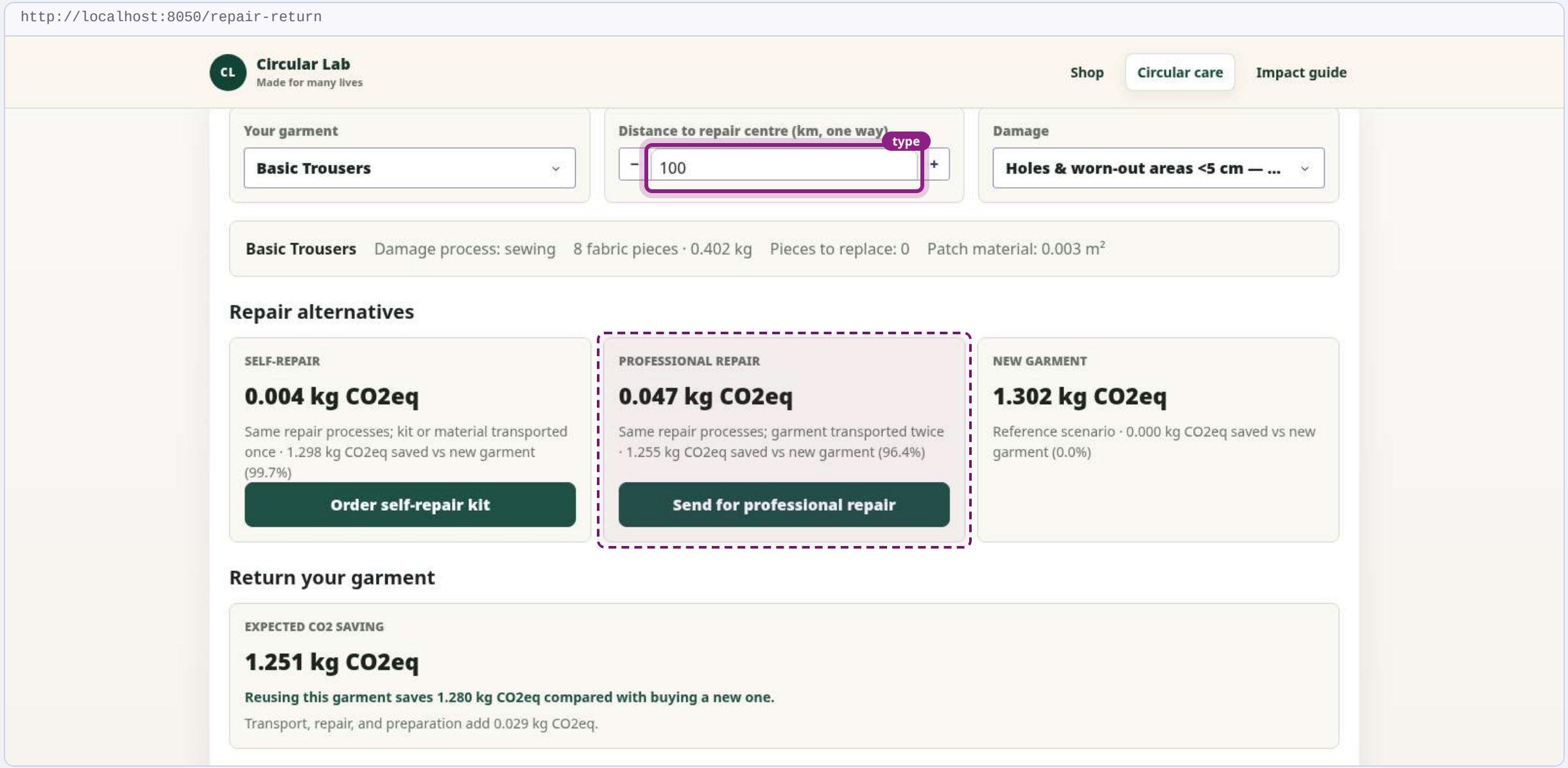
R3 Reuse | R4 Repair | R5 Refurbish | R6 Remanufacture

1. Mia selects her trousers and their damage, a hole under 5 cm: repair is compared with a new garment.



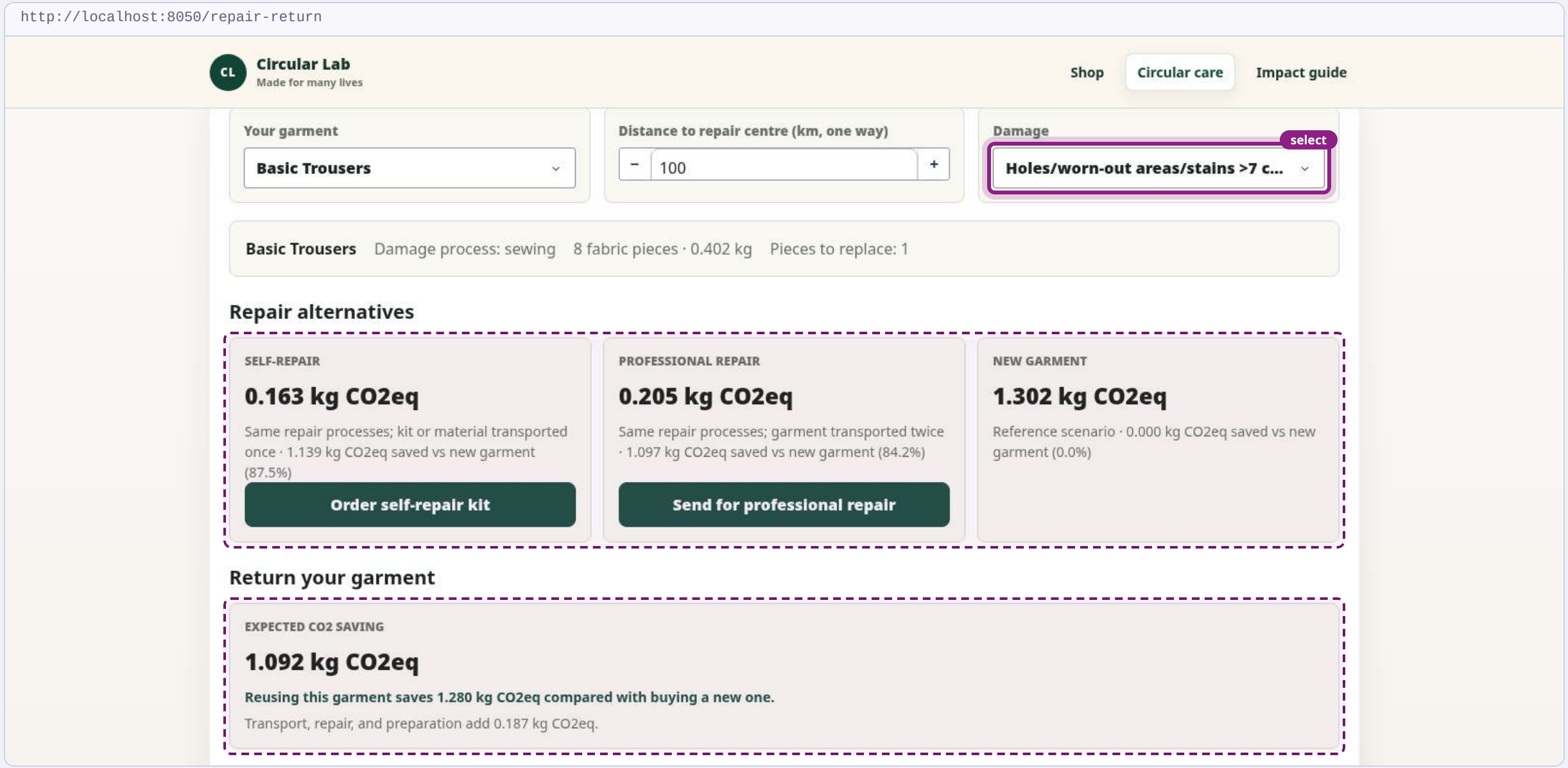
Circular care R3 Reuse R4 Repair R5 Refurbish R6 Remanufacture

2 At 100 km from the repair centre instead of 1000, professional repair drops from 0.452 to **0.047 kg CO2eq**.



Circular care R3 Reuse R4 Repair R5 Refurbish R6 Remanufacture

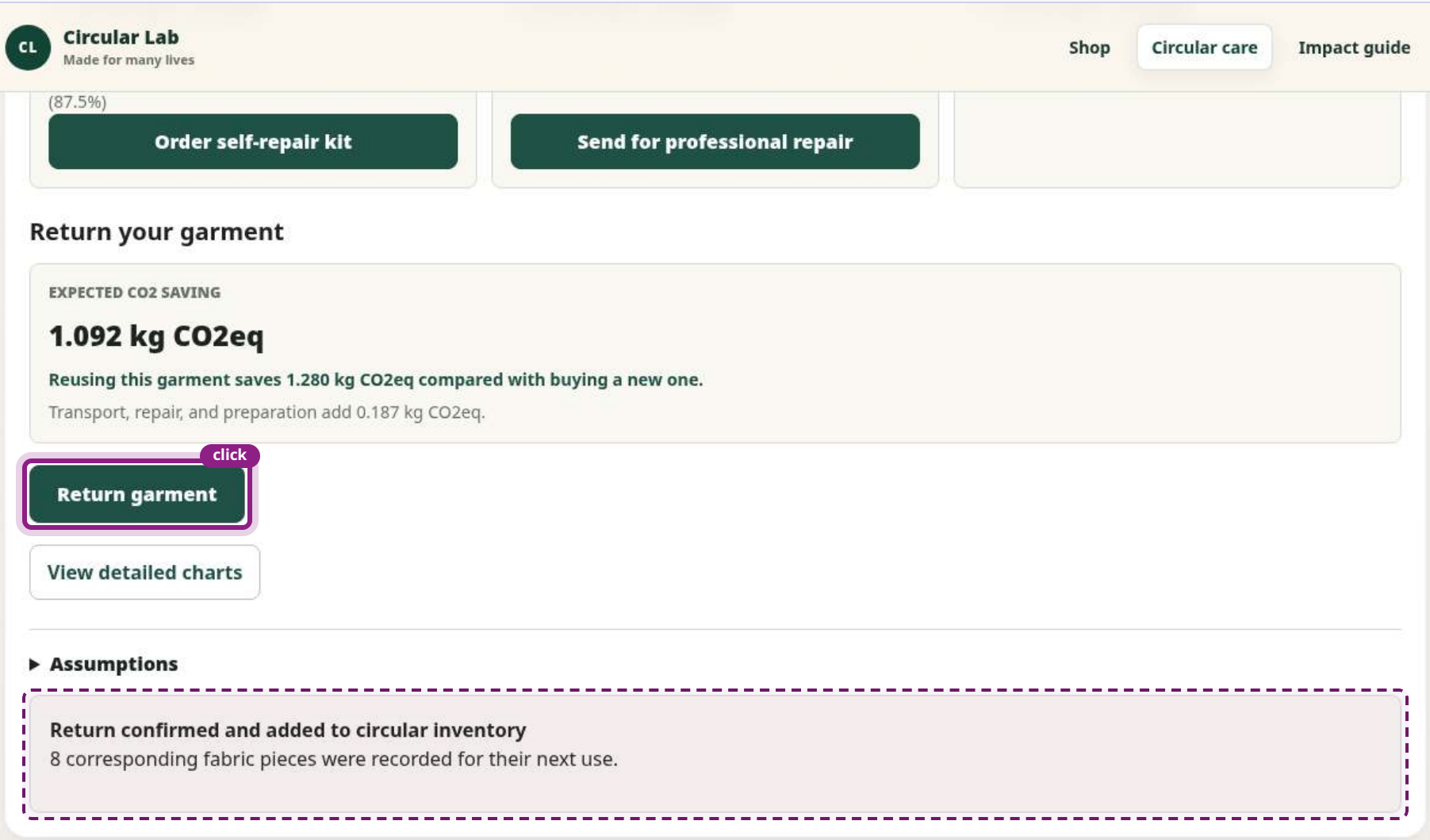
3 A damage over 7 cm needs a new panel: repairs cost more, and a return saves **1.092** instead of 1.251 kg CO2eq.



Circular care R3 Reuse R4 Repair R5 Refurbish R6 Remanufacture

4 Mia returns the trousers: their **8 fabric pieces** are recorded in the circular inventory for their next use.

http://localhost:8050/repair-return



Circular care

R3 Reuse | R4 Repair | R5 Refurbish | R6 Remanufacture

All mentioned prices are fictional

5 Back in the shop, her trousers are offered as pre-loved at **50 % off**: an incentive for customers to return.

http://localhost:8050/garment/1

CL **Circular Lab**
Made for many lives

Shop | Circular care | Impact guide



SELECTED MATERIAL
hemp

ESTIMATED IMPACT
1.312 kg CO2eq

Order garment

50% OFF · ONE AVAILABLE

WHOLE-GARMENT ALTERNATIVE

Give a pre-loved garment its next life

Professionally prepared and ready to wear, with no new fabric required.

CHF 123.35 ~~CHF 246.70~~

Choosing the pre-loved garment avoids new-fabric emissions.

CO2 SAVED	SECOND-LIFE GARMENT	EQUIVALENT NEW
1.307 kg (99.6%)	0.005 kg CO2eq	1.312 kg CO2eq

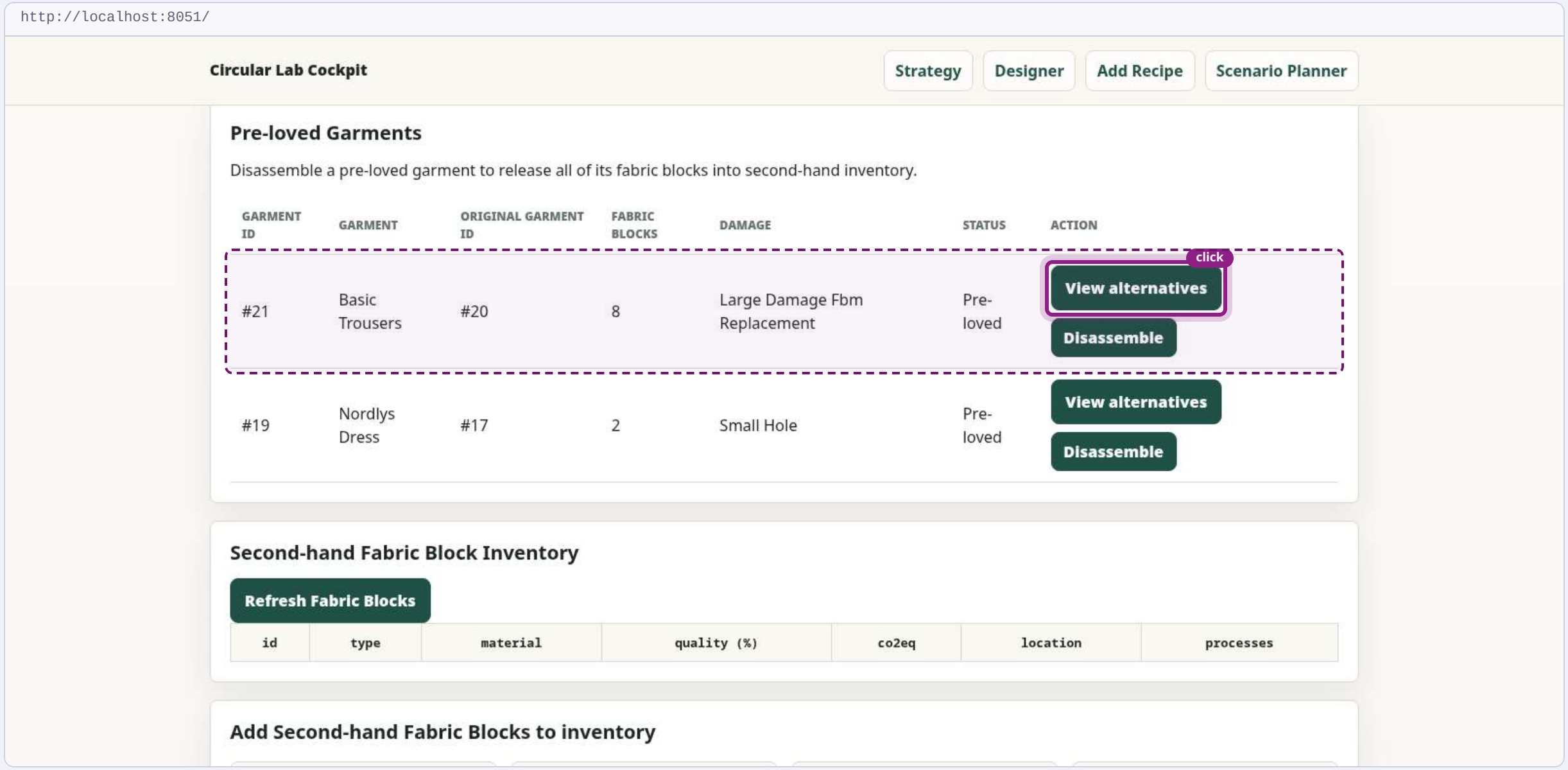
Order pre-loved garment

HOW IT IS MADE

Fabric block inventory

R3 Reuse R4 Repair R5 Refurbish R6 Remanufacture

6 In the cockpit, Mia's return is listed as pre-loved garment **#21**: **View alternatives**.



Fabric block inventory

R3 Reuse | R4 Repair | R5 Refurbish | R6 Remanufacture

7 Reuse the whole garment or recover its fabric blocks? Both net out at about **0.89 kg CO2eq** saved.

http://localhost:8051/

Circular Lab Cockpit

Strategy | Designer | Add Recipe | Scenario Planner

Transport distance (km, one way)

− 1000 +

Electricity factor

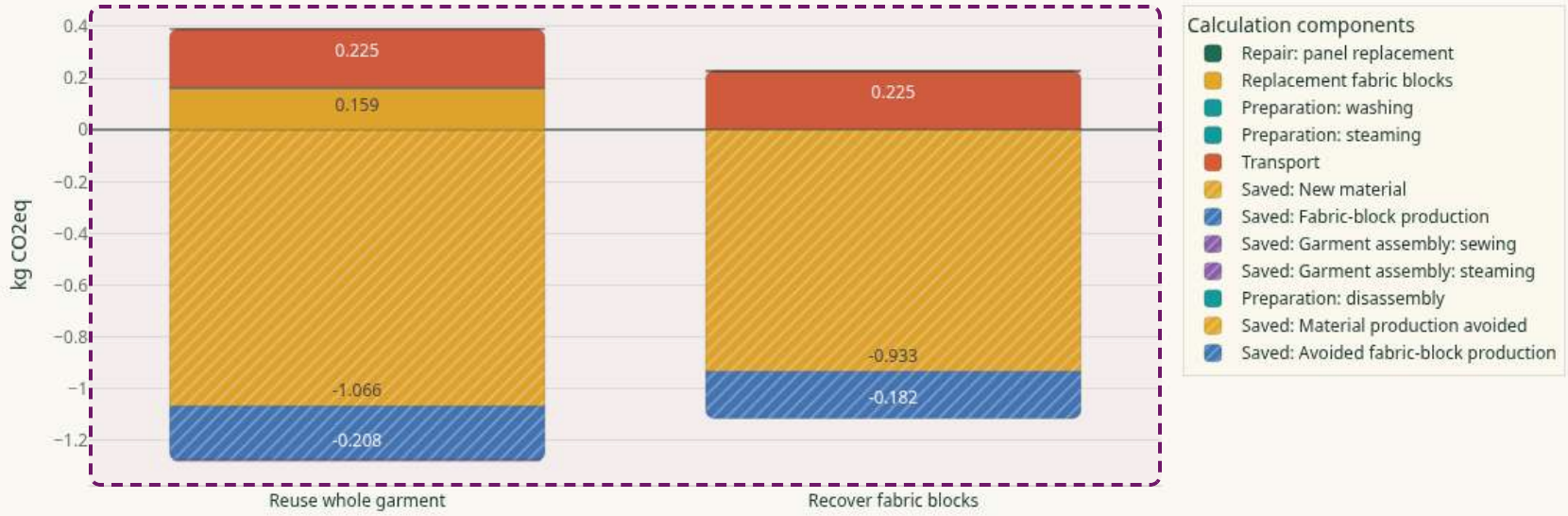
Use higher electricity emission factor

1.253 kg CO2eq/kWh (activity 6566).

Basic Trousers #21 Damage: Holes/worn-out areas/stains >7 cm or deformed shape 8 fabric blocks Current electricity factor

Positive sections are return emissions; negative sections are the new-garment production CO2 saved. Transport is shown only as a return emission.

Return alternatives: impact (+) and CO2 saved (-)



Fabric block inventory

R3 Reuse | R4 Repair | R5 Refurbish | R6 Remanufacture

8 Isolating garment assembly in the legend shows why: it saves under **0.01 kg CO2eq**, so rebuilding is almost loss-less.

http://localhost:8051/

Circular Lab Cockpit

Strategy | Designer | Add Recipe | Scenario Planner

Transport distance (km, one way)

− 1000 +

Electricity factor

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Fabric block inventory

R3 Reuse | R4 Repair | R5 Refurbish | R6 Remanufacture

9 With a higher electricity emission factor, reusing the whole garment saves about **0.16 kg CO2eq** more.

http://localhost:8051/

Circular Lab Cockpit

Strategy | Designer | Add Recipe | Scenario Planner

Transport distance (km, one way)

− 1000 +

Electricity factor

toggle

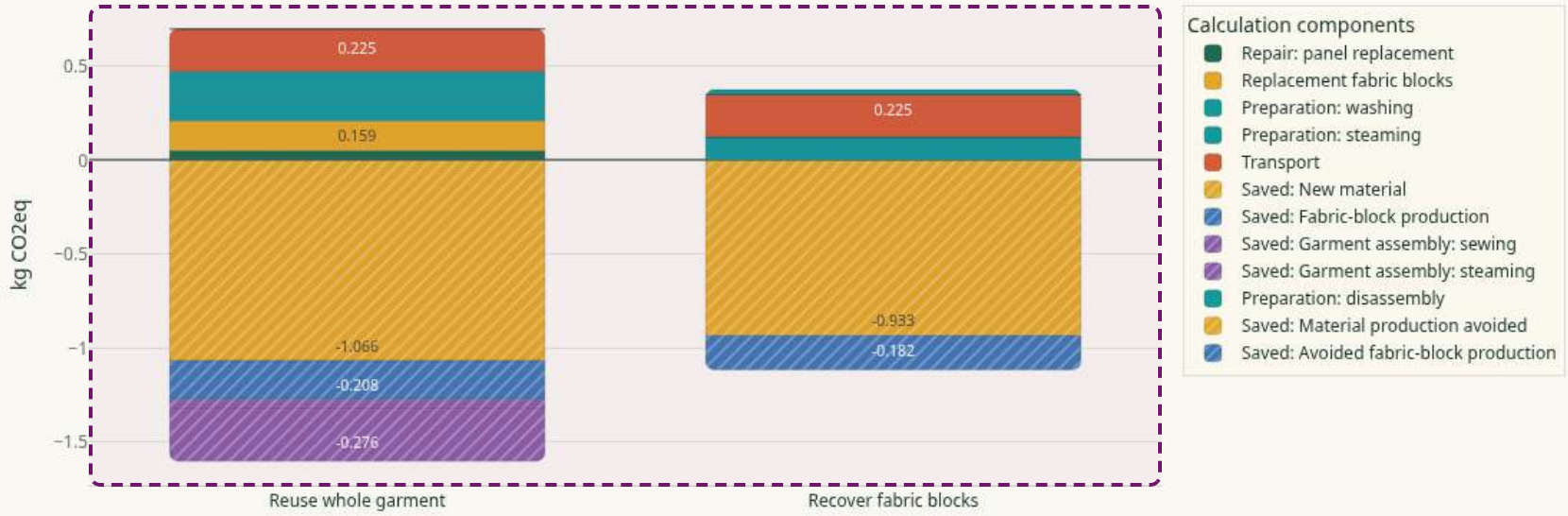
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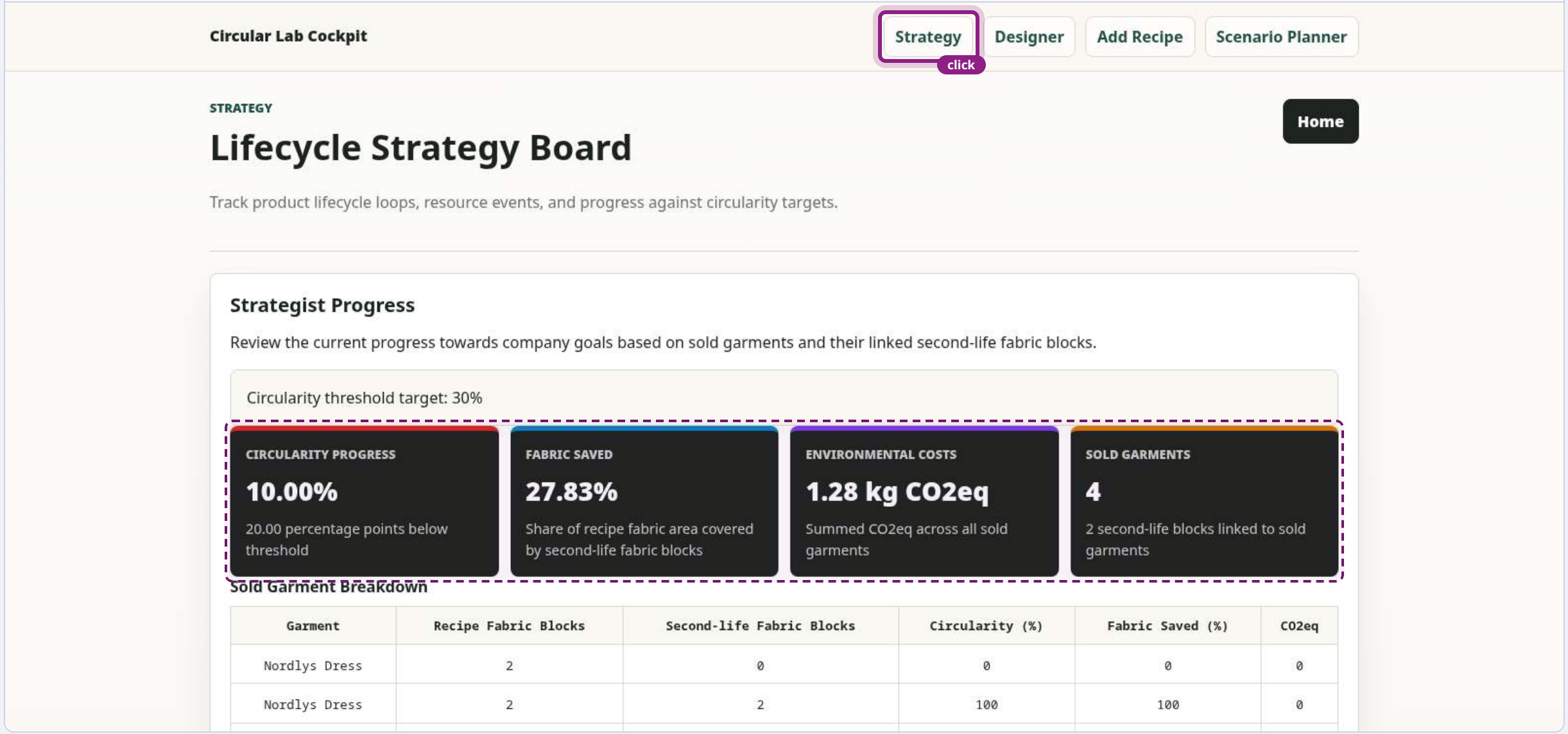


Strategist

R3 Reuse | R4 Repair | R5 Refurbish | R6 Remanufacture

10 Under **Strategy**, the brand tracks its goals: circularity is at 10 %, 20 points below the 30 % target.

http://localhost:8051/dashboard

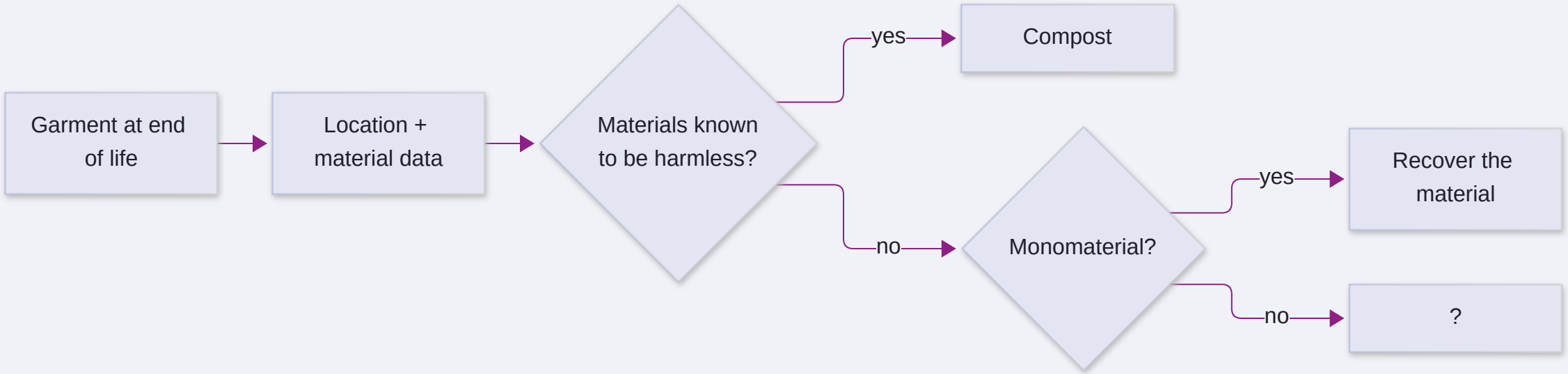


3 - End of Life

R8 Recycle R9 Recover planned

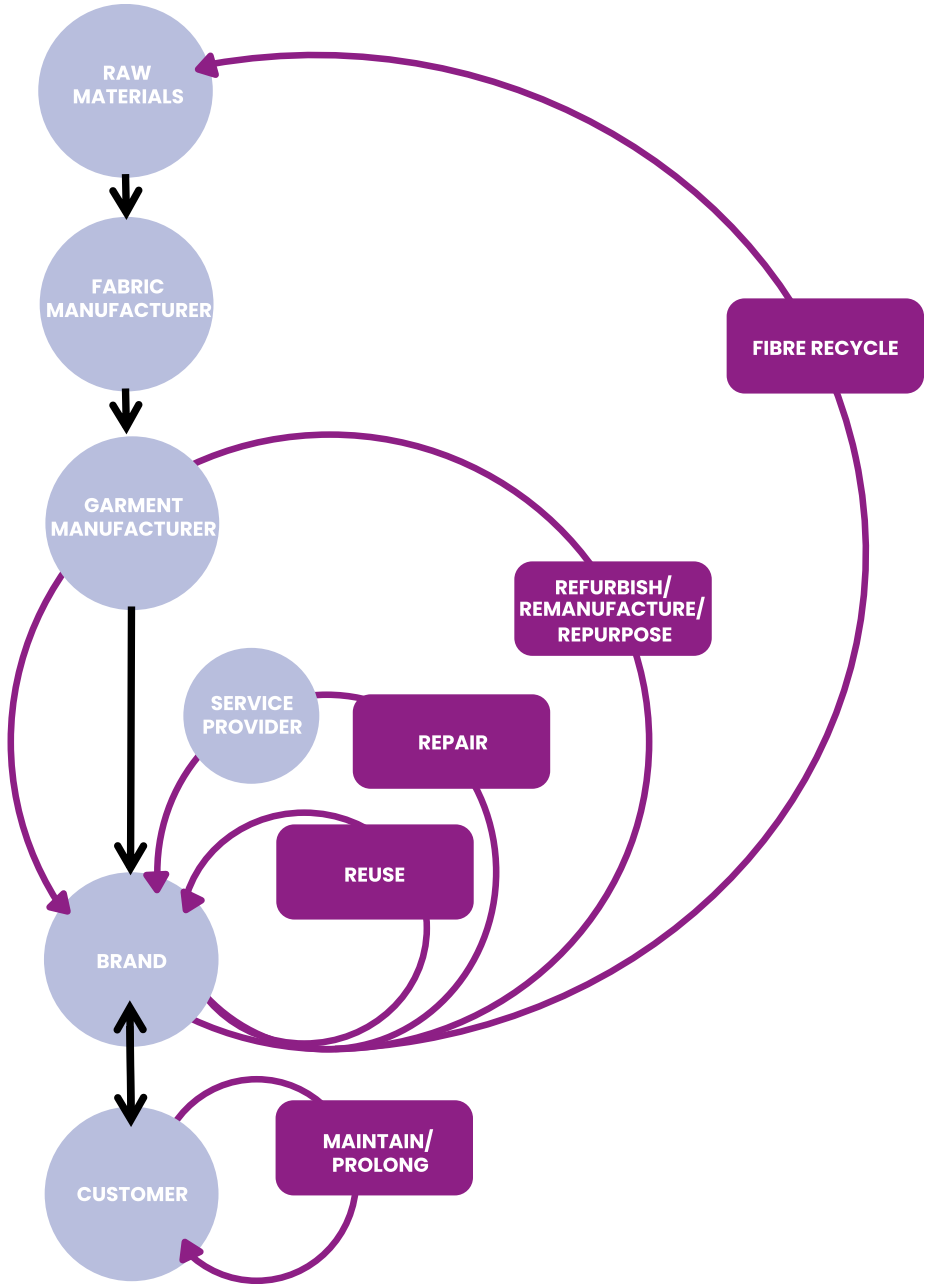
What are the alternatives if R3-R7 are not applicable anymore?

End of Life — future vision for CEIS



In the circular economy, running in circles suddenly makes sense.

CEIS shows how.
To producers and consumers.



Questions?